

Zhiheng Li

UI Engineer | Interaction Designer | Front-End Engineer

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PROFESSIONAL SUMMARY

UI Engineer and Interaction Designer with 15+ years building consumer-facing and B2B digital experiences at the intersection of design and front-end engineering, including 10+ years of hands-on React, HTML5, CSS3, JavaScript (ES6+), and TypeScript development.

Design systems owner: built and maintained multi-brand reusable component libraries, design tokens, and Storybook documentation adopted across distributed product teams, cutting UI build time and eliminating visual drift between design and production.

Fluent in Figma as a working tool, not a hand-off artifact - reads variables, tokens, auto layout, and specs directly from the design file, builds interactive prototypes, and translates design concepts into pixel-accurate, production-ready code.

Deep accessibility practice: WCAG 2.1 AA and Section 508 compliance, semantic HTML, ARIA authoring patterns, keyboard and focus management, screen reader validation (NVDA, VoiceOver), and automated a11y gates (axe-core, Playwright) in CI.

Data visualization and interactive UI specialist - D3.js, Recharts, and Chart.js dashboards plus HTML5 Canvas, WebGL, and Three.js experiences, with motion built in Framer Motion, GSAP, and Lottie.

Headless CMS integration across Sitecore XM Cloud (JSS, Edge Delivery Services), Adobe Experience Manager, and Contentful, wiring structured content into React and Next.js rendering strategies (SSR / SSG / ISR) for SEO and Core Web Vitals (LCP, TTFB, CLS, INP).

Applies AI-powered design and development workflows (Figma AI, v0, GitHub Copilot, Cursor, Claude, ChatGPT) for rapid conceiving, design-to-code generation, component scaffolding, and automated visual and accessibility review.

Domain depth in data-driven B2B environments - commercial real estate and data center infrastructure (Equinix, a global REIT), FinTech, banking, payments, and e-commerce; works cross-functionally with designers, content teams, engineers, and stakeholders, and leads design reviews and design-system governance.

TECHNICAL SKILLS

Design & Prototyping: Figma (Auto Layout, Variables, Variants, Components, Dev Mode), Figma Prototyping, FigJam, Adobe XD, Sketch, wireframing, interaction design, motion design, visual design, responsive and mobile-first design, user flows, design critique

Design Systems: Design tokens (W3C DTCG, Style Dictionary), Storybook, component libraries, atomic design, theming and dark mode, Radix UI, shadcn/ui, Material UI, Chakra UI, Tailwind CSS, design system documentation and governance

Front-End / Frontend Engineering: React 18/19, React Hooks, Next.js (App Router, Server Components, SSR/SSG/ISR), TypeScript, JavaScript (ES6+), HTML5, CSS3, SASS/SCSS, CSS Modules, styled-components, component-based architecture, Redux, Zustand, React Query, React Native

Accessibility: WCAG 2.0 / 2.1 / 2.2 AA, Section 508, ADA, ARIA authoring practices, semantic HTML, keyboard navigation and focus management, color contrast, axe-core, Lighthouse, screen reader testing (NVDA, JAWS, VoiceOver)

Data Visualization: D3.js, Recharts, Chart.js, Visx, AG Grid, SVG, HTML5 Canvas, WebGL, Three.js, enterprise dashboards, real-time charting, interactive data storytelling

Headless CMS & Content: Sitecore XM Cloud, Sitecore JSS, Sitecore Edge Delivery Services, Adobe Experience Manager (Content Fragments, Sling Models, ClientLibs), Contentful, Strapi, content modeling, structured content and omni-channel delivery

APIs & Data: REST, GraphQL, Apollo Client, React Query, WebSockets, JSON, Backend-for-Frontend (BFF) with Node.js and Express, API orchestration

Animation & Interaction: Framer Motion, GSAP, Lottie, CSS animations and transitions, React Native Reanimated, micro-interactions, gesture handling, scroll-driven animation

AI-Assisted Design & Dev: Figma AI, v0, GitHub Copilot, Cursor, Claude, ChatGPT, prompt engineering, AI design-to-code workflows, LLM-powered UI (streaming responses, progressive rendering)

Testing & Tooling: Jest, React Testing Library, Playwright, Cypress, visual regression testing, Chromatic, Vite, Webpack, Turborepo, npm/Yarn, Git, GitHub Actions, GitLab CI, Jira, Agile/Scrum

Performance: Core Web Vitals (LCP, TTFB, CLS, INP), code splitting, lazy loading, memoization, virtualization, selective hydration, image optimization, bundle analysis, caching strategies

PROFESSIONAL EXPERIENCE

UI Engineer / Front-End Architect (Design Systems & Accessibility)

Apple – Sunnyvale, CA Dec 2024 – July 2026

Project: AI-powered documentation platform unifying Apple's API and Markdown documentation on Docusaurus, serving internal engineering and external developer audiences.

Owned the platform's design system end to end - defined design tokens for color, type, spacing, elevation, and motion in Figma Variables, exported them through Style Dictionary, and consumed them in React so a single token change propagated to every surface without a code rewrite.

Built and documented 60+ reusable, accessible React + TypeScript components in Storybook with usage guidelines, do/don't examples, props tables, and live a11y annotations, giving designers and engineers one shared source of truth for the UI.

Translated Figma designs into production-ready code at spec fidelity by pulling variables, auto-layout constraints, and type ramps directly from Dev Mode rather than eyeballing screenshots, removing the design-QA rework loop from the release cycle.

Led UI accessibility strategy to WCAG 2.0/2.1 AA across large-scale consumer and enterprise applications, applying ARIA authoring patterns, semantic landmark structure, keyboard and focus traps, and screen reader validation for inclusive, high-traffic experiences.

Automated accessibility enforcement in CI with axe-core and Playwright, blocking merges on contrast, label, role, and focus-order violations so compliance was verified continuously instead of audited after launch.

Designed interactive documentation and search experiences including data-driven API explorers, live code playgrounds, and contextual navigation, built with React 18, Next.js, and Tailwind CSS.

Built streaming AI-enabled UI components for conversational documentation search - handling partial responses, progressive rendering, retry and fallback states, skeleton loading, and latency-aware interaction design so the interface stayed responsive under model latency.

Integrated headless CMS and structured content pipelines (AEM Content Fragments, Markdown/MDX) into React and Next.js rendering, decoupling content authoring from presentation for omni-channel delivery.

Prototyped concepts rapidly in Figma and AI design tools, then drove them through design reviews with Product, UX, and Content partners, converting critique into shipped interaction patterns.

Optimized Core Web Vitals through code splitting, lazy loading, selective hydration, memoization, and caching, keeping LCP and TTFB within budget on content-heavy documentation pages.

Collaborated cross-functionally with designers, content strategists, backend engineers, and platform teams in an Agile environment, and mentored engineers on component API design, accessibility, and design-system contribution standards.

Environment: React 18, TypeScript, Next.js, Docusaurus, Tailwind CSS, Storybook, Figma, design tokens, Style Dictionary, D3.js, HTML5, CSS3, SCSS, GraphQL, REST, Node.js, Playwright, axe-core, Jest, Webpack, Vite, Git, GitHub Actions, Jira, Agile

UI Engineer / Front-End Architect

Equinix – San Jose, CA Jan 2024 – Dec 2024

Project: B2B enterprise product management platform for a global commercial real estate (REIT) operator of data centers and interconnection services - built from inception to production with quarterly feature delivery.

Integrated Sitecore XM Cloud with React 18 and Next.js using Sitecore JSS and Edge Delivery Services, mapping component-based page composition to SSR/SSG/ISR rendering strategies and improving SEO, LCP, and TTFB on high-traffic marketing and product surfaces.

Designed and shipped the platform's reusable design system and component library with Tailwind CSS and Storybook, establishing token-driven theming, variant APIs, and documentation that reduced release cycle time by roughly 40%.

Built B2B data visualization and dashboard experiences - capacity, utilization, pricing, and interconnection analytics - using D3.js, Recharts, and AG Grid, turning dense operational data into interactive, filterable, drill-down views for enterprise customers.

Led large-scale UI modernization, transforming legacy front-ends into accessible, high-performance SPA, SSR, and mobile-first architectures on React 18, Next.js, Redux, and Tailwind CSS.

Modeled Sitecore and AEM content structures for product metadata, pricing, and documentation, enabling headless content delivery into React and accelerating omni-channel publishing workflows for content teams.

Collaborated with CMS backend teams on Sling Models, ClientLibs, and dispatcher caching to optimize asset delivery, page performance metrics (LCP/TTFB), and multi-region scalability.

Built highly interactive, animation-driven UI and web-based visualizations with HTML5 Canvas, WebGL, and Three.js, applying game-loop rendering patterns and Framer Motion / GSAP motion design for engaging real-time experiences.

Architected Node.js and Express Backend-for-Frontend services aggregating distributed microservices, cutting client-side API calls and simplifying front-end data orchestration.

Designed and optimized REST and GraphQL contracts for type-safe front-end/back-end communication, eliminating over-fetching and under-fetching across distributed applications.

Maintained CI/CD pipelines in GitHub Actions and GitLab CI with Playwright and Cypress suites, accessibility validation, visual regression gates, and deployment automation, reducing production defects by roughly 30%.

Ran cross-browser automation across Chromium, Firefox, and WebKit to guarantee consistent rendering, interaction behavior, and accessibility compliance before release.

Applied front-end security and performance practices including CSP, caching, lazy loading, selective hydration, code splitting, and virtualization, with observability via Prometheus and Grafana.

Provided technical leadership across distributed cross-functional teams - architecture decisions, frontend standards, design reviews, code reviews, and engineer mentoring - partnering with product, UX, backend, and DevOps to define UI architecture and delivery strategy.

Environment: React 18, TypeScript, Next.js, Sitecore XM Cloud, Sitecore JSS, AEM, Tailwind CSS, Storybook, design tokens, Figma, D3.js, Recharts, AG Grid, Three.js, WebGL, HTML5 Canvas, Framer Motion, GSAP, GraphQL, REST, Apollo Client, Node.js, Express, Playwright, Cypress, Jest, Webpack, GitHub Actions, GitLab CI, Git, Jira, Agile

Senior UI Engineer / Front-End Lead

eBay – San Jose, CA Feb 2023 – Dec 2023

Project: E-commerce platform for dealer and consumer vehicle service booking, spanning discovery, scheduling, and transaction management.

Designed and built the end-to-end booking interaction model - multi-step scheduling flows, availability calendars, inline validation, optimistic state, and error recovery - in React 18, Next.js, and TypeScript for high-volume transactional traffic.

Built modular React, Next.js, and Tailwind CSS component libraries integrated with AEM editable templates and Experience Fragments, enabling content teams to compose pages from governed, design-system-approved components.

Developed advanced data-grid and visualization interfaces with AG Grid, D3.js, and Recharts for dealer-facing operational dashboards, supporting large datasets with virtualization, server-side pagination, and interactive filtering.

Automated accessibility testing, WCAG validation, and visual regression testing with Playwright, catching UI inconsistencies and a11y regressions before they reached production.

Leveraged Playwright network interception and API mocking to validate complex front-end workflows independent of backend availability, improving test stability and developer velocity.

Integrated GraphQL APIs with Apollo Client, implementing normalized caching, pagination, optimistic updates, and subscriptions for real-time, responsive interfaces.

Designed AI-driven UI patterns including confidence scoring, explainability indicators, streaming responses, retry handling, and feedback loops, improving transparency and user trust in AI-assisted features.

Applied responsible-AI principles at the UX layer - consent, privacy, explainability, accessibility, and clear labeling of AI-generated versus human-authored content.

Implemented AEM dispatcher caching, front-end optimization, and API orchestration, improving TTFB and LCP while reducing redundant CMS and API requests.

Built and deployed CI/CD pipelines with Tekton, Azure DevOps, GitHub Actions, and GitLab CI, reducing time to market by roughly 40% through automated build, test, and release workflows.

Implemented strict TypeScript typing across components, APIs, and shared libraries using generics, utility types, and inference, reducing runtime errors and improving maintainability.

Environment: React 18, TypeScript, Next.js, Tailwind CSS, AEM, GraphQL, Apollo Client, REST, D3.js, Recharts, AG Grid, HTML5, CSS3, SCSS, Node.js, Playwright, Cypress, Jest, Tekton, Azure DevOps, GitHub Actions, Webpack, Git, Jira, Agile

Senior UI Engineer / Front-End Lead

Wells Fargo – Fremont, CA Dec 2021 – Feb 2023

Project: FinTech loan-processing platform enabling loan officers to analyze, update, and report on applications across multiple banks and customer portfolios.

Designed and built the FlexLoan application experience from scratch - owning the interaction design, visual design, and front-end implementation of a multi-step personal loan application with progressive disclosure, contextual help, and inline eligibility feedback.

Delivered pixel-accurate UI from design specs using a reusable React component library, keeping typography, spacing, and color consistent with the enterprise design system across every screen.

Owned WCAG 2.1 AA and ADA compliance for consumer financial products - semantic structure, ARIA attributes, keyboard operability, error identification, and screen reader validation on regulated, audit-reviewed flows.

Built data visualization and reporting interfaces for loan portfolio analytics with D3.js and Chart.js, giving processors interactive views of pipeline, status distribution, and risk indicators.

Delivered secure FinTech experiences on a regulated banking platform, implementing PCI-DSS-aligned data handling, KYC/AML-aware lending workflows, JWT/OAuth authentication, and audit-friendly compliant UI flows.

Developed reusable, scalable React components integrated with AEM, enabling efficient content management and fast-loading front-end experiences.

Led the team across React development, unit testing, BDD testing, test-data management, CI/CD, multi-environment testing, and accessibility testing.

Drove adoption of Cypress and Playwright for automated testing, stabilizing flaky suites through auto-wait, custom retries, and trace-viewer analysis.

Built AI-enabled document and knowledge interfaces supporting Markdown rendering, semantic search, summarization, and contextual navigation across large content platforms.

Partnered with Product, UX, and Visual Designers to translate concepts into intuitive, accessible, production-ready interfaces, and evaluated AI-driven UI performance for latency, fallback states, and scalability under load.

Optimized React application performance with SSR/SSG in Next.js, caching strategies, lazy loading, code splitting, and efficient API access for SEO-friendly, high-performance delivery.

Built cross-platform mobile experiences in React Native with native iOS (Swift, Objective-C) and Android (Kotlin, Java) module integration, and managed complex state with Redux and Context API.

Environment: React, TypeScript, Next.js, React Native, Redux, Tailwind CSS, HTML5, CSS3, SCSS, Bootstrap, AEM, D3.js, Chart.js, REST, JWT/OAuth, Cypress, Playwright, Jest, Jenkins, Webpack, npm, Git, Jira, Agile

Senior Front-End Developer

Comcast – Philadelphia, PA Sept 2019 – Nov 2021

Project: Migration of legacy Xfinity consumer web properties to a modern React platform, with the build and deploy pipeline decoupled to accelerate the development lifecycle.

Created reusable React components consuming JSON over REST APIs to generate dynamic content across external-facing consumer websites.

Contributed to the migration of legacy systems onto the Xfinity React platform, improving scalability and usability for customer-facing applications.

Decoupled a monolithic architecture into separate front-end and back-end delivery pipelines, accelerating development and testing for high-traffic consumer platforms.

Developed higher-order components for user interaction tracking and analytics reporting, enabling data-driven product decisions by business stakeholders.

Wrote unit tests for React components reaching 99.6% code coverage, raising reliability of customer-facing applications.

Resolved cross-browser compatibility issues across Safari, Internet Explorer, Firefox, and Chrome for consistent behavior across the consumer base.

Built functional prototype pages from UI mockups and ran UI reviews with UI architects and business units to deliver user-centered product sites.

Monitored production incidents with Kibana and Elasticsearch, analyzing logs and troubleshooting issues in external-facing web applications.

Implemented supporting RESTful services in ASP.NET Core for front-end/back-end communication, and collaborated in Agile/Scrum teams with developers, testers, and product owners.

Environment: React, JavaScript (ES6+), HTML5, CSS3, SCSS, Redux, REST, .NET Core, C#, Kibana, Elasticsearch, Jest, Webpack, npm, Git, Jira, Agile

Senior Front-End Consultant

Hero Digital – Iselin, NJ July 2017 – Sept 2019

Project: Replacement of a legacy JSP banking system with a React-based platform for U.S. Bank, including seamless integration and 15 new customer-facing features.

Created reusable React components consuming JSON payloads from Java services to generate dynamic, data-driven page content.

Maintained accessibility on every component and validated with screen readers, ensuring compliant experiences for banking customers.

Architected SCSS file structures for a large-scale application and maintained themed stylesheets for distinct business lines - an early design-system approach to shared styling primitives.

Integrated Adobe Experience Manager with a third-party content platform using customized CSS3 configuration to surface curated social media content as an interactive Photo Booth experience for the GreenDot credit card.

Designed and built semantic HTML5 modal forms with Bootstrap and jQuery, persisting user input and pre-populating downstream steps to reduce drop-off in multi-page flows.

Implemented AES-256 encrypted authentication handshakes and JWT token exchange for secure RESTful communication with backend services.

Wrote unit tests for React components reaching 99.6% code coverage, and used Redux Promise middleware for asynchronous data retrieval from RESTful services.

Integrated Google APIs to build autocomplete address pre-fill, and produced mock JSON payloads to develop and test UI independently of backend readiness.

Built fully responsive single-page applications with Bootstrap, CSS3, and HTML5, and resolved cross-browser issues across Safari, Internet Explorer, Firefox, and Chrome.

Environment: React, Redux, JavaScript (ES6+), HTML5, CSS3, SCSS, Bootstrap, jQuery, JSON, JWT, AEM, REST, Jest, Webpack, npm, Git, Jira, Agile

Web Developer / UI Developer

Dell – New Jersey June 2014 – July 2017

Project: Responsive marketing and promotional web experiences plus Oracle Responsys email template systems supporting global promotional campaigns.

Built responsive, cross-client HTML and CSS email template systems for Oracle Responsys, creating a reusable modular template library that let marketing teams assemble campaigns without hand-coding each send.

Developed responsive marketing pages with HTML5, CSS3, SASS, JavaScript, jQuery, and Bootstrap, translating creative comps into production markup at spec fidelity.

Created interactive data visualizations with D3.js to present campaign performance and product comparison data to business stakeholders.

Implemented AJAX-driven dynamic content and JSON-based data binding to personalize promotional experiences by segment.

Maintained accessible markup and validated with screen readers, and ensured cross-browser consistency across Safari, Internet Explorer, Firefox, and Chrome.

Built prototype pages from creative mockups and reviewed them with UI architects and business units, working in Agile/Scrum delivery.

Environment: HTML5, CSS3, SASS, JavaScript, jQuery, Bootstrap, D3.js, AJAX, JSON, XML, Oracle Responsys, MySQL, Git, Agile

UI Developer

Vertex Inc. – King of Prussia, PA Sept 2013 – Sept 2014

Project: Healthcare benefits portal where members access plan information, health history, and provider search with ratings-based doctor and hospital discovery.

Developed responsive, interactive web pages with HTML5, CSS3, JavaScript, React, Bootstrap, Node.js, and AJAX, communicating with backend services over RESTful APIs.

Built single-page applications with React, ES6, Redux, and Webpack, constructing a modern component-driven front-end architecture.

Created reusable React components and implemented the Redux pattern to maintain predictable, consistently updated application state.

Implemented drag-and-drop interaction for plan and provider comparison, plus forms, routing, events, and animations across the member experience.

Configured Webpack, Babel, and Gulp for transpilation and build tooling, including TypeScript to JavaScript compilation.

Built responsive grid-based layouts for mobile platforms with Bootstrap, and consumed backend JSON to drive DOM and CSS updates in the UI.

Developed Node.js applications on MVC frameworks and managed client-side dependencies with Bower, Grunt, and npm.

Participated in Agile delivery with two-week sprints, daily scrums, and pair programming.

Environment: React, Redux, React Router, Node.js, MongoDB, HTML5, CSS3, JavaScript, TypeScript, jQuery, Bootstrap, REST, AJAX, JSON, ES6, Webpack, Babel, Grunt, npm, GitHub, Jenkins, Jira, Agile

Front-End Developer

JetBlue – New Jersey June 2010 – May 2013

Project: Consumer-facing airline web properties covering promotional campaigns, booking support pages, and customer self-service interfaces.

Developed cross-browser, cross-platform HTML, CSS, and JavaScript to match design specifications for single-page layouts following coding standards.

Created optimized graphics and application interfaces with HTML and CSS for consumer-facing airline properties.

Used jQuery to connect front-end components to JavaScript behavior, adding dynamic interaction and adaptive fluid layouts with Bootstrap.

Implemented AJAX to extend site capability without full page reloads.

Built reusable, modular front-end pages with HTML, CSS, JavaScript, jQuery, responsive web design, and Bootstrap.

Created accessible, table-less screen layouts following W3C standards, and included form validation and panel layouts.

Debugged with Firebug and IE Developer Toolbar, manipulating the DOM to resolve rendering and behavior differences across Safari, Internet Explorer, Firefox, and Chrome.

Developed supporting internal services in Java and wrote SQL queries and stored procedures.

Worked in Agile/Scrum, collaborating with the development team on QA defect resolution and using Git for version control.

Environment: HTML, CSS, JavaScript, jQuery, AJAX, Bootstrap, D3.js, JSON, XML, SASS, Java, MySQL, Git, Agile

EDUCATION

Master of Science, Electrical Engineering

Lehigh University, Bethlehem, PA, USA

Bachelor of Engineering, Industrial Automation

Shanxi University, Shanxi, China